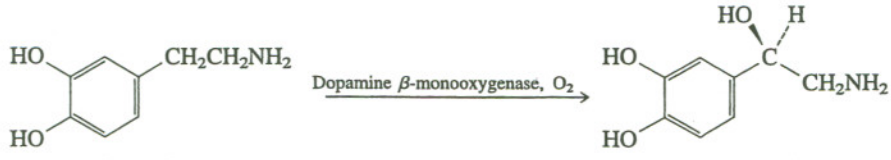
Organic Cume
December 6, 2002

The following problems were taken from the undergraduate organic textbook, Vollhardt & Schore, 4th edition. Answer any **six** of the problems.

57. The enzymatic introduction of a functional group into a biologically important molecule is not only specific with regard to the location at which the reaction occurs in the molecule (see Chapter 4, Problem 41), but also usually specific in the stereochemistry obtained. The biosynthesis of epinephrine first requires that a hydroxy group be introduced specifically to produce (−)-norepinephrine from the achiral substrate dopamine. (The completion of the synthesis of epinephrine will be presented in Problem 63 of Chapter 9.) Only the (−) enantiomer is functional in the appropriate physiological manner, so the synthesis must be highly stereoselective.



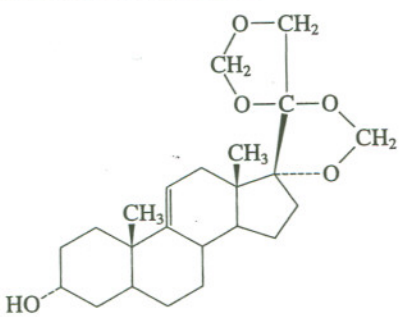
Dopamine **(−)-Norepinephrine**

(a) Is the configuration of (−)-norepinephrine *R* or *S*? **(b)** In the absence of an enzyme, would the transition states of a radical oxidation leading to (−)- and (+)-norepinephrine be of equal or unequal energy? What term describes the relation between these transition states? **(c)** In your own words, describe how the enzyme must affect the energy of these transition states to favor production of the (−) enantiomer. Does the enzyme have to be chiral or can it be achiral?

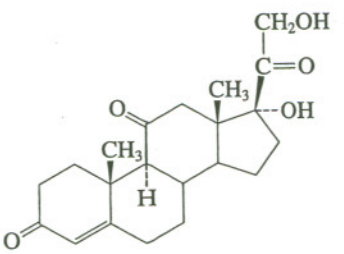
44. Iodoalkanes are readily prepared from the corresponding chloro compounds by S_N2 reaction with sodium iodide in propanone (acetone). This particular procedure is especially useful because the inorganic by-product, sodium chloride, is insoluble in acetone; its precipitation drives the equilibrium in the desired direction. Thus, it is not necessary to use excess NaI, and the process goes to completion in a very short time. Because of its great convenience, this method is named after its developer (the Finkelstein reaction). In an attempt to synthesize optically pure (*R*)-2-iodoheptane, a student prepared a solution of (*S*)-2-chloroheptane in acetone. In order to ensure success, he added excess sodium iodide and allowed the mixture to stir over the weekend. His yield of 2-iodoheptane was high, but, to his dismay, he found that his product was racemic. Explain.

28. Sulfuryl chloride (SO_2Cl_2, see margin for structure) is a liquid reagent that may be used for chlorinations of alkanes as a substitute for gaseous elemental chlorine. Propose a mechanism for chlorination of CH_4 using sulfuryl chloride. (**Hint:** Follow the usual model for a radical chain process, substituting SO_2Cl_2 for Cl_2 where appropriate.)

52. Cortisone is an important steroidal anti-inflammatory agent. Cortisone can be synthesized efficiently from the alkene shown here.

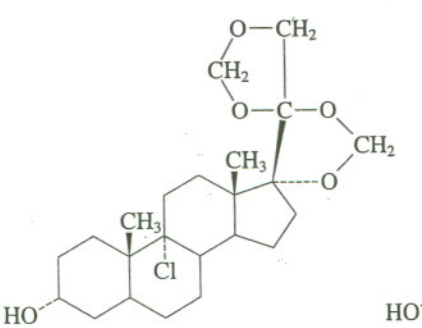


Alkene

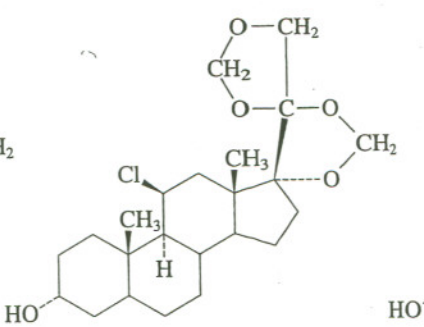


Cortisone

Of the following three chlorinated compounds, two give reasonable yields of the alkene shown above by E2 elimination with base, but one does not. Which one does not work well, and why? What does it give during attempted E2 elimination?



A



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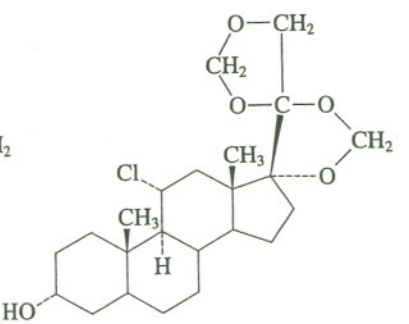
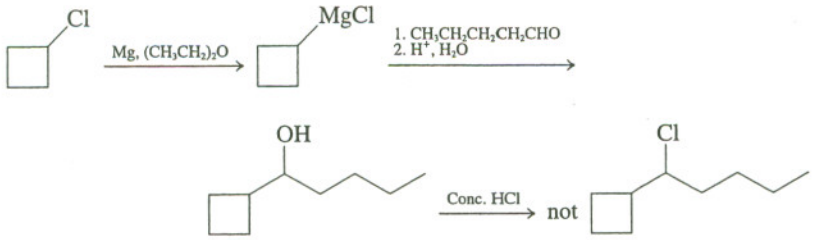
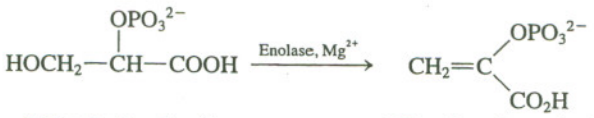


C

54. In an attempt to make 1-chloro-1-cyclobutylpentane, the following reaction sequence was employed. The actual product isolated, however, was not the desired molecule but an isomer of it. Suggest a structure for the product and give a mechanistic explanation for its formation. (**Hint:** See Chapter Integration Problem 9-23.)

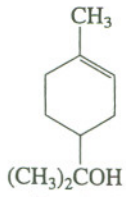


60. Sugars, being polyhydroxylic compounds (Chapter 24), undergo reactions characteristic of alcohols. In one of the later steps in glycolysis (the metabolism of glucose), one of the glucose metabolites with a remaining hydroxy group, 2-phosphoglyceric acid, is converted into 2-phosphoenolpyruvic acid. This reaction is catalyzed by the enzyme enolase in the presence of a Lewis acid such as Mg^{2+}. **(a)** How would you classify this reaction? **(b)** What is the possible role of the Lewis acidic metal ion?



2-Phosphoglyceric acid **2-Phosphoenolpyruvic acid**

58. Treatment of α-terpineol (Chapter 10, Problem 48) with aqueous mercuric acetate followed by sodium borohydride reduction leads predominantly to an isomer of the starting compound ($C_{10}H_{18}O$) instead of a hydration product. This isomer is the chief component in oil of eucalyptus and, appropriately enough, is called eucalyptol. It is popularly used as a flavoring for otherwise foul-tasting medicines because of its pleasant spicy taste and aroma. Deduce a structure for eucalyptol on the basis of sensible mechanistic chemistry and the following proton-decoupled ^{13}C NMR data. (**Hint:** The IR spectrum shows nothing between 1600 and 1800 cm $^{-1}$ or between 3200 and 3700 cm^{-1}!)

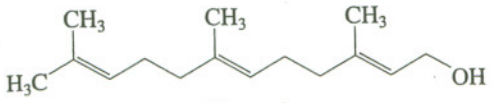


α-Terpineol

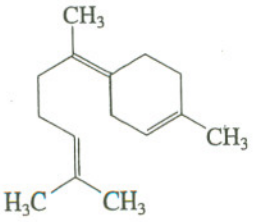
1. $Hg(OCCH_3)_2$, H_2O
2. $NaBH_4$, H_2O

→ eucalyptol, ($C_{10}H_{18}O$) ^{13}C NMR: δ = 22.8, 27.5, 28.8, 31.5, 32.9, 69.6, and 73.5 ppm

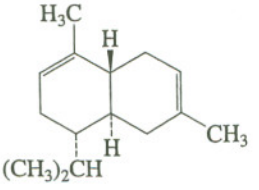
64. Farnesol is a molecule that makes flowers smell good (lilacs, for instance). Treatment with hot concentrated H_2SO_4 converts farnesol first into bisabolene and finally into cadinene, a compound of the essential oils of junipers and cedars. Propose detailed mechanisms for these conversions.



Farnesol



Bisabolene

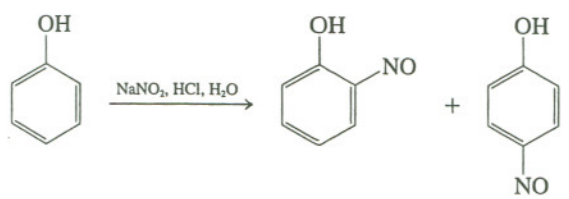


Cadinene

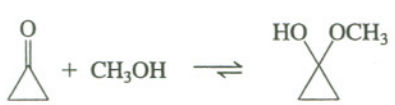
47. **(a)** 3-Phenylpropanoyl chloride, $C_6H_5CH_2CH_2COCl$, reacts with $AlCl_3$ to give a single product with the formula C_9H_8O and an ^1H NMR spectrum with signals at $\delta = 2.53$ (t, $J = 8$ Hz, 2 H), 3.02 (t, $J = 8$ Hz, 2 H), and 7.2–7.7 (m, 4 H) ppm. Propose a structure and a mechanism for the formation of this product.

(b) The product of the process described in (a) is subjected to the following reaction sequence: (1) $NaBH_4$, CH_3CH_2OH; (2) Conc. H_2SO_4, 100°C; (3) H_2, Pd–C, CH_3CH_2OH. The final product exhibits five resonance lines in its proton-decoupled ^{13}C NMR spectrum. What is the structure of the substance formed after each of the steps in this sequence?

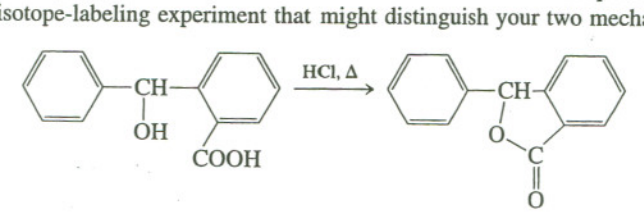
49. Typical conditions for nitrosation are illustrated in the following equation. Propose a detailed mechanism for this reaction.



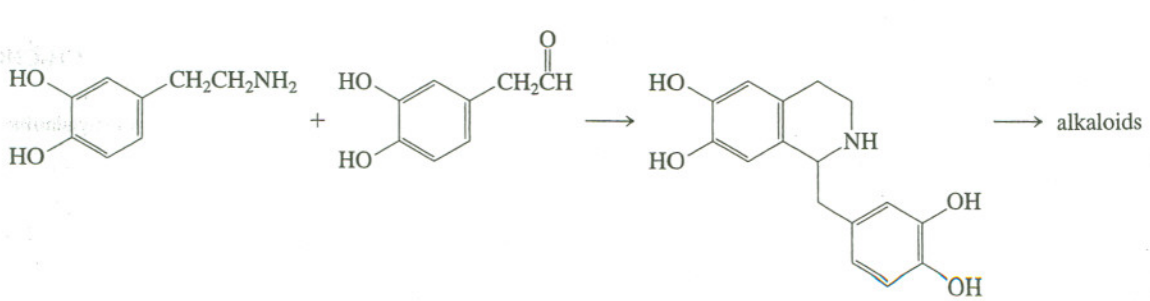
48. Explain the fact that, although hemiacetal formation between methanol and cyclohexanone is thermodynamically disfavored, addition of methanol to cyclopropanone goes essentially to completion:



44. Propose *two* possible mechanisms for the following reaction. (**Hint:** Consider the possible sites of protonation in the molecule and the mechanistic consequences of each.) Devise an isotope-labeling experiment that might distinguish your two mechanisms.



52. Many alkaloids are synthesized in nature from a precursor molecule called norlaudanosoline, which in turn appears to be derived from the condensation of amine A with aldehyde B. Formulate a mechanism for this transformation. Note that a carbon–carbon bond is formed in the process. Name a reaction presented in this chapter that is closely related to this carbon–carbon bond formation.



A **B** **Norlaudanosoline**